Chemistry

Class 10 (English Medium)

Solved Exercises — Sindh Textbook Board

8 lessons — 221 questions

1. کیمیائي توازن

Question: T1. Write down the forward and reverse reactions for the following: N2 + O2 <=> 2NO; 2SO2 + O2 <=> 2SO3; COCl2 <=> CO + Cl2.

Answer: (i) Forward: N2 + O2 -> 2NO; reverse: 2NO -> N2 + O2.

(ii) Forward: 2SO2 + O2 -> 2SO3; reverse: 2SO3 -> 2SO2 + O2.

(iii) Forward: COCl2 -> CO + Cl2; reverse: CO + Cl2 -> COCl2.

Question: T2. Define the term active mass.

Answer: Active mass is the molar concentration of a reacting substance. Its unit is mol dm^-3 and it is written in square brackets, for example [H2].

Question: T3. Figure out the coefficients for the given hypothetical reaction: 9X + Y3 <=> 3X3Y.

Answer: The coefficients in the reaction are 9, 1 and 3. From them the Kc expression is: Kc = [X3Y]^3 / [X]^9 [Y3].

Question: T4. Write down the Kc expression for the given reactions: (i) S + O2 <=> SO2 (ii) SO2 + NO2 <=> NO + SO3 (iii) NH4Cl <=> NH3 + HCl.

Note: the book's worked examples include every species, as in its CO2 + H2 example which also includes H2O.

Answer: (i) Kc = [SO2] / [S] [O2].

(ii) Kc = [NO] [SO3] / [SO2] [NO2].

(iii) Kc = [NH3] [HCl] / [NH4Cl].

Question: T5. The value of Kc for the reaction H2 + I2 <=> 2HI at 717 K is

48. At a particular instant the concentrations of H2, I2 and HI are found to be 0.2, 0.2 and 0.6 mol dm^-3 respectively. Calculate the reaction quotient for the given reaction and predict the direction of the reaction.

Answer: Qc = [HI]^2 / [H2] [I2] = 0.6^2 / 0.2 × 0.2 = 0.36 / 0.04 = 9. Qc, that is 9, is less than Kc, that is 48, so the net reaction proceeds in the forward direction and more HI is formed.

Question: T6. Match each of the following statements with the appropriate diagram: A = Small Kc indicates that the reaction mixture mostly contains reactants. B = Moderate Kc indicates that the reaction mixture contains equal amounts of reactant and product. C = Large Kc indicates that the reaction mixture mostly contains product. Diagrams: (i) diagram in which reactant particles are in the majority. (ii) diagram in which both are equal. (iii) diagram in which product particles are in the majority.

Answer: A - (i), B - (ii), C - (iii).

A1. Which one of the following statements is false about dynamic equilibrium?

(a) It takes place in a closed container.

(b) Concentration of reactants and products are not changed.

(c) Rate of forward reaction is equal to rate of reverse reaction.

✓ (d) Equilibrium cannot be disturbed by any external stress.

A2. Consider the following reaction and indicate which of the following best describes the equilibrium constant expression Kc: 4NH3 + 5O2 <=> 4NO + 6H2O.

✓ (a) Kc = [NO]^4 [H2O]^6 / [NH3]^4 [O2]^5

(b) Kc = [NH3]^4 [O2]^5 / [NO]^4 [H2O]^6

(c) Kc = [NH3] [O2] / [H2O] [NO]

(d) Kc = [4NO] [6H2O] / [4NH3] [5O2]

A3. A reaction which never goes to completion is known as a reversible

reaction. A reversible reaction is represented by:

(a) Dotted lines

(b) Single arrow

✓ (c) Double arrow

(d) Double straight line.

A4. When the magnitude of Kc is small, it indicates:

✓ (a) Reaction mixture contains most of the reactant

(b) Reaction mixture contains most of the product

(c) Reaction mixture contains almost equal amounts of reactant and product

(d) Reaction goes to completion.

A5. For which system does the equilibrium constant Kc have units of concentration?

✓ (a) N2 gas + 3H2 gas <=> 2NH3 gas

(b) N2 gas + O2 gas <=> 2NO gas

(c) H2 gas + I2 gas <=> 2HI gas

(d) CO2 gas + H2 gas <=> CO gas + H2O liquid.

A6. What is the unit of Kc for the reaction N2 + O2 <=> 2NO?

(a) mol dm^-3

(b) mol^-2 dm^6

(c) mol^-1 dm^3

✓ (d) No unit.

A7. The system is stable in equilibrium when:

✓ (a) Qc = Kc

(b) Qc > Kc

(c) Qc < Kc

(d) None of these.

Question: A8. Qc can be defined as: (a) Ratio of product and reactants (b)

Ratio of molar concentration of product and reactant at a specific time (c) Ratio of molar concentration of product and molar volume of reactant (d) Ratio of molar concentration of product and reactant raised to the power of the coefficient.

Answer: (d) Ratio of molar concentration of product and reactant raised to the power of the coefficient.

A9. Which of the following represents the backward reaction? Three diagrams are given: in (i) blue particles are in the majority and red ones are few; in (ii) green particles are in the majority and there is one golden particle; in (iii) red particles are in the majority and blue ones are few.

A10. The value of Kc increases when:

(a) Product is less

(b) Product is more

(c) Reactant is more

(d) Reactant equals product.

Question: B1. Define chemical equilibrium with an example.

Answer: Chemical equilibrium is the state in which, in a reversible reaction in a closed container, the rate of the forward reaction becomes equal to the rate of the reverse reaction and the concentrations of the reactants and products stay constant. Example: H2 + I2 <=> 2HI.

Question: B2. Why is chemical equilibrium dynamic?

Answer: Because at equilibrium the two reactions do not stop; the forward and reverse reactions keep going at equal rates, so no change is seen from outside, but internally the reaction continues.

Question: B3. When writing an equation, how is a reversible reaction distinguished from an irreversible reaction?

Answer: A reversible reaction is written with a double arrow <=>, which shows it proceeds in both directions. An irreversible reaction is written

with a single arrow ->, which shows it proceeds in only one direction.

Question: B4. Write an equilibrium equation with monoatomic carbon and a molecule of oxygen as reactants and carbon monoxide as product.

Answer: 2C + O2 <=> 2CO. Two carbon atoms and one molecule of oxygen form two molecules of carbon monoxide, so the equation is balanced.

Question: B5. Outline the characteristics of a reversible reaction.

Answer: A reversible reaction takes place in a closed container; it never goes to completion; both the forward and reverse reactions proceed; equilibrium can be reached from either side; it can be disturbed by external stress; at equilibrium the rates of both reactions are equal and the concentrations stay constant.

Question: B6. Distinguish between reversible and irreversible reactions.

Answer: A reversible reaction proceeds in both directions, never goes to completion, is shown by a double arrow, and can establish equilibrium. An irreversible reaction proceeds in only one direction, goes to completion, is shown by a single arrow, and does not establish equilibrium.

Question: B7. State the law of mass action. How is the active mass represented?

Answer: At any instant, the rate of a chemical reaction is directly proportional to the product of the active masses of the reactants at that instant. Active mass means molar concentration, which is written in square brackets, for example [H2], and its unit is mol dm^-3.

Question: B8. Why may the equilibrium constant have a unit or not? Justify with an example.

Answer: Whether Kc has a unit depends on whether the numbers of moles on the two sides are equal or not. If the moles are equal the units cancel, for example in N2 + O2 <=> 2NO Kc has no unit. If the moles are not

equal Kc has a unit, for example in N2 + 3H2 <=> 2NH3 the unit of Kc is mol^-2 dm^6.

Question: B9. How can the direction of a reaction be predicted if Kc is known to you?

Answer: Qc is compared with Kc. If Qc < Kc the net reaction proceeds forward. If Qc > Kc the net reaction proceeds in reverse. If Qc = Kc the system is in equilibrium and no net change occurs.

Question: B10. Write the equilibrium constant expression for the following equations: (i) N2 + 2O2 <=> 2NO2 (ii) N2 + 3H2 <=> 2NH3 (iii) H2 + Br2 <=> 2HBr.

Answer: (i) Kc = [NO2]^2 / [N2] [O2]^2.

(ii) Kc = [NH3]^2 / [N2] [H2]^3.

(iii) Kc = [HBr]^2 / [H2] [Br2].

Question: C1. Describe dynamic equilibrium with two examples.

Answer: In dynamic equilibrium the forward and reverse reactions keep going at equal rates, so the concentrations stay constant. Example 1: H2 + I2 <=> 2HI, in which hydrogen and iodine combine to form hydrogen iodide and HI breaks apart again. Example 2: N2 + 3H2 <=> 2NH3, in which nitrogen and hydrogen form ammonia and ammonia breaks apart again. In both cases no change is seen outwardly, but internally both reactions continue.

Question: C2. State the law of mass action. Derive an expression for the equilibrium constant.

Answer: Law: at any instant, the rate of a chemical reaction is directly proportional to the product of the active masses of the reactants at that instant. For the general reaction aA + bB <=> cC + dD, forward rate = kf

[A]^a [B]^b and reverse rate = kr [C]^c [D]^d. At equilibrium the two rates are equal, so kf [A]^a [B]^b = kr [C]^c [D]^d, which gives kf / kr = [C]^c [D]^d / [A]^a [B]^b = Kc.

Question: C3. Describe the characteristics of the equilibrium constant.

Answer: Kc works only at equilibrium; it expresses the equilibrium concentrations of reactants and products in mol dm^-3; it is independent of the initial concentrations; it changes with temperature, and temperature is the single factor affecting its value; if Kc is greater than 1 the reaction is in the forward direction, and if it is less than 1 it is in the reverse direction; Kc tells both the direction and the extent of the reaction.

Question: C4. How can you predict the following stages of a reaction by comparing the values of Kc and Qc: (i) Net reaction proceeds in the forward direction. (ii) Net reaction proceeds in the reverse direction.

Answer: (i) When Qc < Kc the net reaction proceeds forward so that the product concentration increases.

(ii) When Qc > Kc the net reaction proceeds in reverse so that the product concentration decreases and more reactants form. When Qc = Kc the system is in equilibrium.

Question: C5. Predict which system at equilibrium will contain the maximum amount of product and which system will contain the maximum amount of reactant: (a) 2CO2 <=> 2CO + O2, Kc at 927 C = 3.1 × 10^-18 mol dm^-3 (b) 2O3 <=> 3O2, Kc at 298 K = 5.9 × 10^55 mol dm^-3.

Answer: In (a) Kc is very small, so at equilibrium there will be mostly reactant, that is CO2, and very little product. In (b) Kc is very large, so at equilibrium there will be mostly product, that is O2, and very little reactant.

Question: D1. Dinitrogen tetraoxide N2O4 decomposes into nitrogen

dioxide NO2 in a reversible reaction. Derive the equilibrium constant expression for the decomposition reaction. Also interpret the unit of Kc for the balanced chemical reversible reaction.

Answer: Reaction: N2O4 <=> 2NO2. Kc = [NO2]^2 / [N2O4]. Unit: the top has mol dm^-3 squared and the bottom has mol dm^-3, so the unit of Kc is mol dm^-3.

Question: D2. PCl5, PCl3 and Cl2 are at equilibrium at 500 K in a closed container and their concentrations are 0.8 × 10^-3, 1.2 × 10^-3 and 1.2 × 10^-3 mol dm^-3 respectively. Calculate the value of Kc for the reaction along with its unit.

Answer: Reaction: PCl5 <=> PCl3 + Cl2. Kc = [PCl3] [Cl2] / [PCl5] = 1.2 × 10^-3 × 1.2 × 10^-3 / 0.8 × 10^-3 = 1.44 × 10^-6 / 0.8 × 10^-3 = 1.8 × 10^-3 mol dm^-3.

Question: D3. The value of Kc for the reaction 2HI <=> H2 + I2 is 1 × 10^-4. At a given temperature, the molar concentration of the reaction mixture is HI = 2 × 10^-5 mol dm^-3, H2 = 1 × 10^-5 mol dm^-3 and I2 = 1 × 10^-5 mol dm^-3. Predict the direction of the reaction.

Answer: Qc = [H2] [I2] / [HI]^2 = 1 × 10^-5 × 1 × 10^-5 / (2 × 10^-5)^2 = 1 × 10^-10 / 4 × 10^-10 = 0.25. Qc, that is 0.25, is greater than Kc, that is 1 × 10^-4, so the net reaction proceeds in the reverse direction and more HI is formed.

2. تیزاب، بنیاد لوٹ

Question: T1. Why is Arrhenius theory only applicable to aqueous solutions?

Answer: Because according to Arrhenius theory an acid is one that gives H+ in aqueous solution and a base is one that gives OH- in aqueous solution. Without water these ions are not formed, so the theory applies only to aqueous solutions.

Question: T2. Write down the conjugate acid and conjugate base of the following reactants: (I) HCl + H2O (II) H2SO4 + H2O (III) NH4+ + H2O.

Answer: (I) The conjugate base of HCl is Cl- and the conjugate acid of H2O is H3O+. (II) The conjugate base of H2SO4 is HSO4- and the conjugate acid of H2O is H3O+. (III) The conjugate base of NH4+ is NH3 and the conjugate acid of H2O is H3O+.

Question: T3. Why is pure water considered a weak electrolyte?

Answer: Because pure water ionizes only very slightly: in it [H+] = [OH-] = 1 × 10^-7 mol dm^-3, so very few ions are present to conduct electricity.

Question: T4. What is the pH value of the following? Lemon juice, soda water, black coffee, milk, washing soda.

Answer: Lemon juice 2.3, soda water 3.8, black coffee 5.0, milk 6.3 to 6.6, washing soda 12.0.

Question: T5. Which of the following are salts? HCl, NaCl, NaOH, KOH, K2SO4, KNO3, HNO3, BaCl2.

Answer: Salts are: NaCl, K2SO4, KNO3, BaCl2. Not salts: HCl and HNO3 are acids; NaOH and KOH are bases.

Question: T6. List down the types of salts.

Answer: (i) Acidic salts, whose pH is less than 7, for example NH4Cl and

NaHSO4.

(ii) Basic salts, whose pH is more than 7, for example CH3COONa and Na2CO3.

(iii) Neutral salts, whose pH is 7, for example NaCl and KCl.

A1. Corrosive effect on skin is caused by:

(a) Acid

(b) Base

(c) Salt

✓ (d) Both a and b.

A2. Preservatives are used to preserve:

(a) Acids

(b) Bases

✓ (c) Food

(d) Water.

A3. Which of the following is not an Arrhenius acid?

(a) HCl

✓ (b) CO2

(c) HNO3

(d) H2SO4.

A4. NH3 can be a base according to:

(a) Arrhenius theory

(b) Bronsted-Lowry theory

(c) Lewis theory

✓ (d) Both b and c.

A5. Which of the following is a Lewis base?

(a) HNO3

(b) CN-

(c) HCl

(d) AlCl3.

A6. A substance that can donate a pair of electrons to form a coordinate covalent bond:

(a) Lewis acid

✓ (b) Lewis base

(c) Bronsted-Lowry acid

(d) Bronsted-Lowry base.

A7. If the pH value is greater than 7, then the solution is:

(a) Acidic

✓ (b) Basic

(c) Amphoteric

(d) Neutral.

A8. Salt among the following is:

(a) HCl

✓ (b) KCl

(c) HNO3

(d) H2SO4.

A9. Substances that react with both acids and bases are called:

(a) Conjugate acids

(b) Conjugate bases

✓ (c) Amphoteric substances

(d) Buffers.

A10. The reaction of an acid and a base to form salt and water is called:

(a) Hydration

✓ (b) Neutralization

(c) Hydrolysis

✓

(d) Both a and c.

Question: B1. Discuss the properties of acids and bases.

Answer: Acids are sour in taste, give H+ in aqueous solution, change blue litmus paper to red, and are corrosive. Bases are bitter in taste, give OH- in aqueous solution, change red litmus paper to blue, and feel soapy and slippery.

Question: B2. Elaborate the Arrhenius concept of acids and bases with a suitable example.

Answer: According to Arrhenius, an acid is one that gives H+ in aqueous solution, for example HCl -> H+ + Cl-. A base is one that gives OH- in aqueous solution, for example NaOH -> Na+ + OH-.

Question: B3. What is the Bronsted-Lowry acid-base theory?

Answer: According to this theory an acid is a proton donor and a base is a proton acceptor. Example: in HCl + NH3 -> NH4+ + Cl-, HCl donates a proton so it is the acid, and NH3 accepts a proton so it is the base.

Question: B4. What are conjugate acid-base pairs? Explain with examples.

Answer: When an acid donates a proton, what remains is its conjugate base, and when a base accepts a proton, what forms is its conjugate acid. Examples: the conjugate base of HCl is Cl-; the conjugate acid of NH3 is NH4+; the conjugate acid of H2O is H3O+ and its conjugate base is OH-.

Question: B5. Classify the following solutions as acidic, basic or neutral: A solution that has [H+] = 1 × 10^-4 mol dm^-3. A solution that has [H+] = 1 × 10^-11 mol dm^-3. A solution that has [OH-] = 1 × 10^-9 mol dm^-3. A solution that has [OH-] = 1 × 10^-3 mol dm^-3.

Answer: [H+] = 1 × 10^-4 means pH 4, so acidic. [H+] = 1 × 10^-11 means pH 11, so basic. [OH-] = 1 × 10^-9 means pOH 9 and pH 5, so acidic. [OH-] = 1 × 10^-3 means pOH 3 and pH 11, so basic.

Question: B6. Elaborate the ionization equation of water.

Answer: Water ionizes slightly: H2O <=> H+ + OH-, or 2H2O <=> H3O+ + OH-. The ionization constant for water is Kw = [H+] [OH-] = 1 × 10^-14 at 25 C. In pure water [H+] = [OH-] = 1 × 10^-7 mol dm^-3, so its pH is 7.

Question: B7. Define the following terms: (a) pH (b) Indicator (c) Neutralization (d) Titration.

Answer: (a) pH is the measure of the hydrogen ion concentration: pH = -log [H+].

(b) An indicator is a weak organic acid or base which changes its colour over a small range of pH.

(c) Neutralization is the reaction between an acid and a base to produce salt and water.

(d) Titration is a technique where a solution of known concentration is used to determine the concentration of an unknown solution.

Question: B8. Define buffers. What is the composition of buffers? Discuss their importance in our daily life.

Answer: A buffer is an aqueous solution which resists a change in its pH against the addition of a small amount of acid or base. The composition of a buffer is a weak acid and its salt, that is its conjugate base, for example CH3COOH and CH3COONa, or a weak base and its salt, that is its conjugate acid, for example NH3 and NH4Cl. In daily life buffers keep the pH of blood and body fluids stable, so that the serum of blood stays at about pH 7.4.

Question: C1. Describe salts, preparation of salts and types of salts.

Answer: A salt is an ionic compound in which the cation comes from the

base and the anion comes from the acid; it is obtained along with water in the neutralization of an acid and a base. Preparation: the action of acids on metals, metal oxides, metal hydroxides, metal carbonates and metal bicarbonates; the action of a base with an acid; the action of a metal with a base. Types: (i) Acidic salts, pH less than 7, for example NH4Cl.

(ii) Basic salts, pH more than 7, for example Na2CO3.

(iii) Neutral salts, pH 7, for example NaCl.

Question: C2. Explain in detail how water ionization is related to the pH of a solution.

Answer: Water ionizes: H2O <=> H+ + OH-, and Kw = [H+] [OH-] = 1 × 10^-14 at 25 C. pH = -log [H+] and pH + pOH = 14. In pure water [H+] = [OH-] = 1 × 10^-7, so the pH is 7 and water is neutral. If H+ increases the pH falls and the solution becomes acidic; if OH- increases the pH rises and the solution becomes basic.

Question: C3. Discuss in detail how different solutions in an aqueous system exhibit an increase in acidity and an increase in basicity on the basis of H+ and OH-.

Answer: In an aqueous system acidity depends on the amount of H+ and basicity on the amount of OH-. If [H+] > [OH-] the solution is acidic and the pH is less than 7; the more H+, the greater the acidity. If [OH-] > [H+] the solution is basic and the pH is more than 7; the more OH-, the greater the basicity. If both are equal the solution is neutral and the pH is 7.

Question: C4. What do you mean by balancing of neutralization reaction with the help of examples?

Answer: Balancing a neutralization reaction means that the H+ coming from the acid and the OH- coming from the base are in equal numbers, so

that water forms and the salt remains. Example: in NaOH + HCl -> NaCl + H2O there is one H+ and one OH-. In 2NaOH + H2SO4 -> Na2SO4 + 2H2O, 2NaOH is written to make two H+ and two OH- equal. In Ca(OH)2 + 2HNO3 -> Ca(NO3)2 + 2H2O the H+ and OH- are also made equal.

Question: C5. Write down the uses of salt in daily life.

Answer: Salts are used as fertilizers in agriculture, for example ammonium chloride and NPK fertilizers. Some salts are used as pesticides, for example copper sulphate. Plaster of Paris is used to make plaster casts for broken bones. Iron sulphate is used in medicine for anaemia. Sodium hydrogen carbonate is used in antacids for stomach acidity. Barium sulphate is used in X-rays of the stomach. Potassium permanganate is used as a disinfectant. Common salt is used in food for taste and preservation.

Question: D1. Calculate the pH of 5M solution of NaOH.

Answer: [OH-] = 5 mol dm^-3. pOH = -log 5 = -0.70. pH = 14 - pOH = 14 - (-0.70) = 14.70.

Question: D2. A solution of H2SO4 has a pH of 1.05. Calculate its pOH and [H+].

Answer: pOH = 14 - pH = 14 - 1.05 = 12.95. [H+] = 10^-pH = 10^-1.05 = 8.9 × 10^-2 mol dm^-3.

Question: D3. The hydrogen ion concentration of a solution is 1 × 10^-8 mol dm^-3. What is the pH of the solution?

Answer: pH = -log [H+] = -log 1 × 10^-8 = 8. The solution is basic.

3. نامياتي کيمسٹري

Question: T1. Why are the number of compounds of Si lesser than that of carbon compounds?

Answer: Because carbon has a much greater power of catenation than silicon. The C-C bond, at 355 kJ mol^-1, is much stronger than the Si-Si bond at 200 kJ mol^-1. On the other hand the Si-O bond, at 452 kJ mol^-1, is stronger than the C-O bond at 351 kJ mol^-1, so silicon is found in nature as silica and silicates and does not form long chains.

Question: T2. What is the rate of reactivity of organic compounds?

Answer: The rate of reaction of organic compounds is slow, because their covalent bonds take time to break and need more energy. In contrast, ionic compounds react fast.

Question: T3. Name the alkanes up to decane.

Answer: Methane CH4, ethane C2H6, propane C3H8, butane C4H10, pentane C5H12, hexane C6H14, heptane C7H16, octane C8H18, nonane C9H20, decane C10H22.

Question: T4. Which type of bonding do unsaturated hydrocarbons possess?

Answer: Unsaturated hydrocarbons possess multiple bonds between carbon atoms: alkenes have a double bond C=C and alkynes have a triple bond, written here as C#C.

Question: T5. Identify the functional group in the given compounds: CH3CH2OH, CH3OCH3, CH3CHO.

Answer: In CH3CH2OH there is -OH, the alcoholic group. In CH3OCH3 there is -O-, the ether linkage. In CH3CHO there is -CHO, the aldehydic group.

Question: T6. If we dip a blue litmus paper in a solution and it changes to

red colour, which functional group does the solution have?

Answer: The solution is acidic, so it has the carboxylic acid group -COOH, which gives the acidic character.

A1. The branch of chemistry which deals with hydrocarbons and their derivatives is known as:

✓ (a) Organic chemistry

(b) Inorganic chemistry

(c) Biochemistry

(d) Physical chemistry.

A2. The general formula for alkanes is:

(a) CnH2n

(b) CnH2n+1

✓ (c) CnH2n+2

(d) CnH2n-2.

A3. Which of the following is an alcohol?

(a) CH3CHO

(b) CH3CH2O CH3

✓ (c) CH3OH

(d) HCOOH.

A4. Which of the following is a saturated hydrocarbon?

(a) CH3CH=CH2

✓ (b) CH3CH2CH3

(c) CH3CCH

(d) CH2=CH-CCH.

A5. The prefix hept stands for carbon atoms:

(a) 2

(b) 5

✓ (c) 7

(d) 9.

A6. The functional group COOH is used for:

(a) Alkynes

(b) Alcohols

(c) Phenols

✓ (d) Carboxylic acids.

A7. Polyethene is:

(a) Oil

(b) Paper

✓ (c) Plastic

(d) Wood.

A8. Acetic acid is obtained from:

(a) Banana

(b) Dates

(c) Garlic

✓ (d) Vinegar.

A9. Alkenes:

(a) Show the same general formula as alkynes.

(b) Have carbon-carbon triple bond.

✓ (c) Have carbon-carbon double bond.

(d) Are saturated hydrocarbons.

A10. CH3-CH2- is which radical?

(a) Methyl

✓ (b) Ethyl

(c) n-propyl

(d) Isopropyl.

Question: B1. Define Vital Force Theory.

Answer: According to the Vital Force Theory, organic compounds can only

be prepared in living tissues by a vital force and cannot be prepared in the laboratory.

Question: B2. Explain how petroleum is a source of organic compounds.

Answer: Petroleum is a complex mixture of hydrocarbons formed underground from dead plants and animals. It is separated by fractional distillation into separate useful fractions, for example petrol, kerosene and diesel, all of which are organic compounds.

Question: B3. Define the functional group. Write the functional groups which contain carbon, hydrogen and oxygen.

Answer: A functional group is an atom or group of atoms that gives a molecule its characteristic properties. The functional groups containing carbon, hydrogen and oxygen are: alcohol -OH, ether -O-, aldehyde -CHO, ketone -CO-, carboxylic acid -COOH and ester -COOR.

Question: B4. Define the alkyl radicals with suitable examples.

Answer: An alkyl radical is formed by removing one hydrogen atom from an alkane. Examples: methyl CH3- from methane CH4, ethyl CH3CH2- from ethane C2H6, propyl C3H7- from propane.

Question: B5. What is a homologous series? Name some common homologous series.

Answer: A homologous series is a group of compounds whose members have the same functional group, the same general formula, and differ from one another by CH2. Common series are: alkanes, alkenes, alkynes, alcohols, carboxylic acids.

Question: B6. Identify the functional groups in the following compounds: (a) CH3-CHO (b) CH3-CH2-CH2-OH (c) CH3-CO-CH3 (d) CH3-COOH (e) CH2=CH-CH3.

Answer: (a) In CH3-CHO the -CHO, aldehyde group.

(b) In CH3-CH2-CH2-OH the -OH, alcohol group.

(c) In CH3-CO-CH3 the -CO-, ketone group.

(d) In CH3-COOH the -COOH, carboxylic acid group.

(e) In CH2=CH-CH3 the C=C, alkene double bond.

Question: B7. Write the condensed and structural formulae of pentane and octane.

Answer: Pentane: molecular formula C5H12, condensed CH3(CH2)3CH3, structural CH3-CH2-CH2-CH2-CH3. Octane: molecular formula C8H18, condensed CH3(CH2)6CH3, structural CH3-CH2-CH2-CH2-CH2-CH2-CH2-CH3.

Question: B8. What is catenation? Give any two examples of catenation of carbon atoms.

Answer: Catenation is the ability of carbon atoms to join with one another through covalent bonds to form long chains or rings. Examples: in ethane two carbon atoms are joined to each other; in pentane five carbon atoms are joined in a long chain; in cyclohexane carbon atoms are joined in a ring.

Question: C1. Give the important characteristics of organic compounds.

Answer: Carbon is an essential element in organic compounds; organic chemistry is the study of hydrocarbons and their derivatives; because of catenation and isomerization of carbon there are millions of compounds; they dissolve in organic solvents; their bonding is weaker than that of ionic compounds, so they have lower melting and boiling points; their rate of reaction is slow; on combustion all give carbon dioxide; the members of a homologous series have the same functional group; many are used in

perfumes, scents, paints, dyes and drugs.

Question: C2. Differentiate between saturated and unsaturated hydrocarbons.

Answer: Saturated hydrocarbons have single bonds between carbon atoms, for example alkanes CnH2n+2. Unsaturated hydrocarbons have double or triple bonds between carbon atoms, for example alkenes CnH2n with C=C and alkynes CnH2n-2 with a triple bond, written here as C#C.

Question: C3. What are the main sources of organic compounds? Mention with special reference to coal, petroleum and natural gas.

Answer: There are two main sources of organic compounds: living things and minerals. From living things, animals give proteins and fats, and plants give carbohydrates, proteins, vitamins and oils. From minerals come coal, petroleum and gas. Coal is formed from trees buried in the earth: wood 40 percent carbon, peat 60, lignite 70 and anthracite 90 percent. Petroleum is a mixture of hydrocarbons formed from dead plants and animals, separated by fractional distillation. Natural gas is mostly methane; in Pakistan it is called Sui gas.

Question: C4. Describe the uses of organic compounds.

Answer: Organic compounds are used as food, as clothing, as houses, as fuel, as medicines and as raw materials. They are used in making perfumes, dyes, soaps, plastics and drugs. The pharmaceutical chemist plays an important role in laboratory analysis, quality assurance and the production of new effective drugs.

Question: C5. Name the alkenes and alkynes having the following formulae: (i) C2H4 (ii) C3H4 (iii) C3H6 (iv) C6H12 (v) C5H8 (vi) C8H16 (vii) C7H12 (viii) C6H10.

Answer: (i) C2H4 ethene, alkene.

(ii) C3H4 propyne, alkyne.

(iii) C3H6 propene, alkene.

(iv) C6H12 hexene, alkene.

(v) C5H8 pentyne, alkyne.

(vi) C8H16 octene, alkene. (vii) C7H12 heptyne, alkyne. (viii) C6H10 hexyne, alkyne.

Question: C6. Define nomenclature and describe the I.U.P.A.C nomenclature rules for alkynes.

Answer: Nomenclature is the system of giving systematic names to compounds. Rules for alkynes: select the longest continuous chain containing the triple bond; number the chain from the end nearest the triple bond; name = prefix for the number of carbons + yne, for example ethyne, propyne; if there are branches, their names and position numbers are written first.

Question: C7. What do you mean by diversity and magnitude of organic compounds?

Answer: Diversity and magnitude means that organic compounds are very many and of very many kinds: more than 10 million organic compounds exist, far more than the compounds of all the other elements. The reasons are catenation, isomerism, the strong covalent bonds of carbon, and multiple bonding.

4. حياتياتي کيمسٹري

Question: T1. What are carbohydrates?

Answer: Carbohydrates are organic compounds containing carbon, hydrogen and oxygen, with the general formula Cx(H2O)y. They contain an aldehyde group -CHO or a ketone group >C=O along with many hydroxyl groups -OH. Examples: glucose, fructose, starch, cellulose.

Question: T2. Differentiate between monosaccharides and oligosaccharides.

Answer: Monosaccharides are simple sugars that cannot be simplified further on hydrolysis, for example glucose and fructose. Oligosaccharides give 2 to 10 monosaccharides on hydrolysis, for example sucrose and lactose.

A1. Which of the following is a carbohydrate?

(a) Protein

(b) Lipid

✓ (c) Carbohydrate

(d) Vitamin.

A2. Rickets are caused by the deficiency of which vitamin?

(a) Vitamin A

✓ (b) Vitamin D

(c) Vitamin B

(d) Vitamin C.

A3. In any living organism, which thing is more important?

(a) Protein

(b) Vitamin

✓ (c) DNA

(d) Nucleic acid.

A4. Which of the following is an aldose?

(a) Glucose

(b) Fructose

(c) Sucrose

✓ (d) Aldose.

A5. What is the largest structural unit of protein?

(a) Amino acid

✓ (b) Protein

(c) Glucose

(d) Cellulose.

A6. Which of the following is a polysaccharide?

(a) Glucose

(b) Fructose

✓ (c) Starch

(d) Lactose.

A7. Which sugar is called milk sugar?

(a) Glucose

(b) Fructose

✓ (c) Lactose

(d) Sucrose.

A8. Protein is which type of compound?

(a) Homo polymer

(b) Hetero polymer

(c) Monomer

✓ (d) Polyamide.

A9. Proteins at ordinary room temperature are:

✓ (a) At ordinary room temperature

(b) At 100 C

(c) At 50 C

(d) At 0 C.

A10. Cotton is composed of what percentage of cellulose?

(a) 75

(b) 65

(c) 85



(d) 95.

Question: B1. What do you know about proteins?

Answer: Proteins are polymers of amino acids, joined to one another by peptide bonds. One protein may contain from 60 to 6000 amino acid molecules. Proteins are the building blocks of the body, catalyze biological reactions as enzymes, and carry oxygen as haemoglobin.

Question: B2. Write a note on the importance of DNA.

Answer: DNA is the store house of information. It stores and transmits hereditary information from parents to children, encodes genetic information and transfers it from generation to generation. Present in the nuclei of cells, it carries the information for making the different organs like heart, liver, eyes, kidneys, legs and head.

Question: B3. Differentiate between fats and oils.

Answer: Fats are solids at room temperature and increase the cholesterol level in the body, for example butter and ghee. Oils are liquids at room temperature, for example sunflower and olive oil. Both are esters of fatty acids with glycerol.

Question: B4. Define polysaccharides. In what way are monosaccharides obtained from polysaccharides by hydrolysis?

Answer: Polysaccharides are carbohydrates which give more than 10 monosaccharides on hydrolysis, for example starch, cellulose and

glycogen. In hydrolysis, reaction with water breaks the polysaccharide apart and releases its monosaccharides, for example the hydrolysis of starch gives glucose.

Question: B5. Define a peptide bond. What type of peptides are there? Which are called di- and tripeptides?

Answer: A peptide bond is the covalent bond that joins two amino acids to each other, between the carboxyl group of one and the amino group of the other. Peptides can be dipeptides, tripeptides and polypeptides. One made of two amino acids is called a dipeptide and one made of three is called a tripeptide.

Question: B6. Write down the uses of lipids.

Answer: Lipids act as transporters of fatty acids and fat-soluble vitamins A, D, E and K in the body. Some lipids reduce the cholesterol level. Fats and oils are used for cooking and frying food. They are used in detergents, soaps, cosmetics, polishes and paints. They activate the enzymes.

Question: B7. What are amino acids? Write their general structure.

Answer: Amino acids are the building blocks of proteins; they have an amino group -NH2 and a carboxyl group -COOH on the same carbon atom. General structure: a central carbon atom joined to H, -NH2, -COOH and a side chain R.

Question: B8. Write down the sources and importance of vitamin D.

Answer: Vitamin D is a fat-soluble vitamin. It is necessary to keep bones and teeth strong, because it helps the body use calcium and phosphorus. Its deficiency causes rickets in children and weakness of bones in adults. Excess of vitamin D in the body causes calcification and pain in bones.

Question: B9. Differentiate between fat-soluble and water-soluble vitamins.

Answer: Fat-soluble vitamins are A, D, E and K; they are stored in the body

and an excess of them can cause harm. Water-soluble vitamins are the B complex and C; they are not stored in the body and an excess of them cannot harm us.

Question: C1. Describe the sources and uses of carbohydrates.

Answer: Sources: monosaccharides in grapes, honey and guava; oligosaccharides in sugarcane, sugar beet, milk, wheat and vegetables; polysaccharides in the cell walls of all land plants, cotton, cereal foods and potato. Uses: a source of energy for survival; they sustain the structure of plants; storage of energy in the form of starch in plants and glucose in mammals; they keep blood sugar levels in check; sucrose as a food additive; cellulose gives food its bulk and fibre and promotes peristalsis in the intestine; cellulose is a raw material in textiles and paper; starch is used in the production of ethanol and in laundries.

Question: C2. Describe the sources and uses of lipids.

Answer: Sources: from animals, marine animals like salmon and whales, butter, ghee and cheese; from plants, sunflower, coconut, ground nuts, corn, cotton seed and olive. Uses: transporters of fatty acids and fat-soluble vitamins; some lipids reduce the cholesterol level; fats and oils for cooking and frying food; in detergents, soaps, cosmetics, polishes and paints; they activate the enzymes; butter and ghee for bakery products and sweets.

Question: C3. What are vitamins? Describe their types and the role of each one.

Answer: Vitamins are accessory growth factors which are needed in small amounts for the growth and health of the body. Types: (i) Fat-soluble vitamins A, D, E and K, which are stored in the body.

(ii) Water-soluble vitamins B complex and C, which are not stored. Vitamin

A is needed for sight, D for bones, E for protection of cells, K for clotting of blood, B complex for energy and nerves, and C for skin and immunity.

Question: C4. What are nucleic acids? Describe RNA and DNA.

Answer: Nucleic acids are macromolecules formed by the polymerization of nucleotides; each nucleotide is composed of a pentose sugar, a phosphate group and a nitrogenous base. They store and transmit hereditary information. DNA is the store house of information; it encodes genetic information and transfers it from generation to generation. RNA receives, decodes, reads and uses the information obtained from DNA.

Question: C5. Describe the vitamin-deficiency diseases.

Answer: Deficiency of vitamin A causes weakness of sight; deficiency of vitamin D causes rickets; deficiency of vitamin B complex causes nerve and energy problems; deficiency of vitamin C causes scurvy; deficiency of vitamin K causes delay in clotting of blood. Excess also causes harm: vitamin A causes irritation and headache, D causes calcification and pain in bones, E causes fatigue and headache, and K causes liver and kidney diseases.

5. ماحولياتي کيمسٹري-I: فضا

Question: T1. What is atmosphere?

Answer: The atmosphere is the blanket of gases around the earth that covers the earth and is held to the earth by gravity. It is necessary for life and controls weather, air and temperature.

Question: T2. From which gases is our atmosphere made up of?

Answer: The atmosphere is mostly made up of nitrogen 78.09 percent and oxygen 20.95 percent, followed by argon 0.93 percent and carbon dioxide 0.03 percent. Besides these there are water vapour, ozone and other small gases.

Question: T3. How would you differentiate between stratosphere and troposphere?

Answer: The troposphere is the lowest layer, from the earth's surface to about 12 kilometres, in which weather occurs and aeroplanes fly, and the temperature falls as you go higher. The stratosphere is the second layer, from 12 to about 50 to 55 kilometres, which contains the ozone layer, where the temperature rises as you go higher and air movement is low.

A1. Second highest layer of Earth's atmosphere is:

(a) Stratosphere

(b) Mesosphere

(c) Troposphere

✓ (d) Thermosphere.

A2. Aeroplanes fly in:

✓ (a) Troposphere

(b) Stratosphere

(c) Mesosphere

(d) Thermosphere.

A3. Atmospheric pressure decreases with the:

(a) Increase in longitude

(b) Decrease in altitude

✓ (c) Increase in altitude

(d) Increase in latitude.

A4. The layer of atmosphere which separates stratosphere and troposphere is known as:

✓ (a) Tropo-pause

(b) Mesopause

(c) Thermopause

(d) Stratopause.

A5. Ozone layer is part of:

(a) Mesosphere

✓ (b) Stratosphere

(c) Thermosphere

(d) Troposphere.

A6. Which is not part of greenhouse gases?

(a) Carbon dioxide

(b) Methane

(c) Nitrous oxide

✓ (d) Oxygen.

A7. Second most abundant constituent of dry air in terms of volume after nitrogen is:

(a) Nitrogen

✓ (b) Oxygen

(c) Carbon dioxide

(d) Helium.

A8. Which of the following is the reason of global warming?

(a) Presence of sulphite

✓ (b) Rise in CO_2 concentration

(c) Oxides of nitrogen

(d) Formation of ozone.

A9. The altitude of stratosphere is:

(a) 40 to 45 km

✓ (b) 50 to 55 km

(c) 60 to 65 km

(d) 70 to 75 km.

A10. Ozone is a gas found in the layer:

(a) Troposphere

(b) Mesosphere

✓ (c) Stratosphere

(d) Exosphere.

Question: B1. Enlist major air pollutants and their sources.

Answer: Major air pollutants and their sources: carbon dioxide, sulphur dioxide and oxides of nitrogen, from vehicles and factories; smoke and dust, from burning of fuel; pesticides and fertilizers, from land; ozone and acid rain, from secondary reactions.

Question: B2. Describe the effects of some air pollutants on human health.

Answer: Air pollutants affect the respiratory systems of both animals and humans. Acid rain and smoke damage the eyes and lungs. Breathing polluted air can cause cough, asthma, lung diseases and allergy.

Question: B3. What is the cause of acid rain?

Answer: The cause of acid rain is sulphur dioxide SO_2 and oxides of nitrogen NO_2 from factories and vehicles, which combine with water vapour in the atmosphere to form acidic compounds and come down with

the rain.

Question: B4. Justify that greenhouse effect leads to global warming.

Answer: Greenhouse gases like CO2, CH4 and water vapour trap the sun's heat in the atmosphere. An increase in these gases traps more heat, which raises the earth's temperature, which is called global warming. Therefore the greenhouse effect leads to global warming.

Question: B5. List down the layers of atmosphere.

Answer: There are 5 main layers of the atmosphere, from the ground upward: troposphere, stratosphere, mesosphere, thermosphere and exosphere.

Question: B6. Write down the effects of acid rain.

Answer: Acid rain affects the respiratory system; it harms the aquatic ecology; it corrodes water pipelines, so heavy metals like iron, lead and copper seep into drinking water; it damages stone and metal buildings and monuments.

Question: B7. Justify that change in altitude changes the temperature of the atmosphere.

Answer: In the troposphere the temperature falls as altitude increases, from 15 C to -56 C, because the air pressure is lower higher up. In the stratosphere the temperature rises as altitude increases, because the ozone layer absorbs radiation from the sun and gives heat. Therefore a change in altitude changes the temperature.

Question: B8. What are primary and secondary air pollutants?

Answer: Primary pollutants come directly from sources, for example CO2 from vehicles and SO2 from factories. Secondary pollutants are formed in the atmosphere by chemical reactions of primary pollutants, for example acid rain and ozone.

Question: C1. Define the atmosphere and explain its composition.

Answer: The atmosphere is the blanket of gases around the earth that covers the earth and is held to the earth by gravity. Composition: the atmosphere is mostly composed of nitrogen 78.09 percent and oxygen 20.95 percent, followed by argon 0.93 percent and carbon dioxide 0.03 percent. Besides these there are water vapour, ozone, helium, methane and other small gases. Around 80 percent of the mass of the atmosphere is in the troposphere.

Question: C2. Differentiate between stratosphere and troposphere.

Answer: The troposphere is the lowest layer, about 11 to 12 kilometres high, containing about 75 to 80 percent of the mass of the atmosphere; the temperature falls from 15 C to -56 C with height; there is much air movement and an active weather system; almost all planes pass through this layer. The stratosphere is the second layer, from 12 to 50 to 55 kilometres, with far less atmosphere; the temperature rises with height; the ozone layer shields the planet from ultraviolet radiation; air movement is low; aeroplanes are not permitted in this layer.

Question: C3. Describe how different air pollutants affect the environment and human health.

Answer: Air pollutants harm both the environment and health. CO2 and other greenhouse gases cause global warming. SO2 and NO2 form acid rain, which damages the aquatic ecology, pipelines, buildings and monuments. Smoke and dust cause respiratory diseases like cough, asthma and lung diseases. Ozone damages the eyes and lungs. Overall, pollutants endanger the lives of living organisms.

Question: C4. Describe global warming.

Answer: Global warming is the continuous rise in the average temperature of the earth. Its cause is the increase in greenhouse gases, which trap the sun's heat in the atmosphere. As a result the weather changes, ice melts,

sea level rises and living organisms are badly affected.

6. ماحولياتي کيمسٹري-II: پاڻي

A1. Which of the following diseases is caused by a virus?

✓ (a) Polio

(b) Cholera

(c) Typhoid

(d) Diarrhoea.

A2. How much of the earth's surface is covered by water?

✓ (a) 70 percent

(b) 60 percent

(c) 80 percent

(d) 50 percent.

A3. Which bond exists between water molecules?

✓ (a) Hydrogen bond between water molecules

(b) Covalent bond

(c) Ionic bond

(d) Coordinate bond.

A4. What causes permanent hardness in water?

✓ (a) MgSO4

(b) calcium bicarbonate

(c) magnesium bicarbonate

(d) sodium chloride.

A5. How much percentage of water is fresh water?

(a) 0.3 percent

✓ (b) 3 percent

(c) 0.2 percent

(d) 2 percent.

A6. Which of the following causes temporary hardness?

(a) CaCl2

(b) MgCl2

(c) CaSO4

✓ (d) calcium bicarbonate and magnesium bicarbonate.

A7. Water is:

✓ (a) Polar solvent

(b) Non-polar solvent

(c) Non-polar acid

(d) Polar acid.

A8. Water is:

(a) Colourless

(b) Odourless

(c) Tasteless

✓ (d) Tasteless.

A9. What removes permanent hardness?

✓ (a) Na2CO3

(b) NaCl

(c) CaCl2

(d) MgCl2.

Question: B1. Describe the composition of water.

Answer: A water molecule is made up of one atom of oxygen and two atoms of hydrogen connected by covalent bonds: H2O. Because of the higher electronegativity of oxygen, the oxygen carries a partial negative charge and the hydrogens carry partial positive charges, so water is a polar molecule.

Question: B2. Define hard and soft water.

Answer: Soft water is water that forms lather easily with soap and contains

very few salts of calcium and magnesium. Hard water is water that does not form lather easily with soap, because it contains dissolved salts of calcium and magnesium.

Question: B3. Define a water pollutant.

Answer: A water pollutant is a substance that mixes in water and makes it unfit for drinking and other uses, for example chemical waste, sewage, pesticides, fertilizers and germs.

Question: B4. Describe the methods to remove temporary hardness of water.

Answer: Temporary hardness is caused by Ca(HCO3)2 and Mg(HCO3)2.

Methods: (i) Boiling: on heating, the bicarbonate breaks down into insoluble carbonate which settles at the bottom.

(ii) Clark's method: adding slaked lime Ca(OH)2 converts the bicarbonate into carbonate which settles at the bottom.

(iii) Adding washing soda Na2CO3.

Question: B5. Write down the names of waterborne diseases.

Answer: Waterborne diseases: diarrhoea, dysentery, cholera, cryptosporidium, fluorosis, hepatitis, hookworm, jaundice and typhoid.

Question: B6. Differentiate between hard and soft water.

Answer: Soft water forms lather easily with soap and contains very few salts of calcium and magnesium. Hard water does not form lather easily with soap, because it contains dissolved chlorides, sulphates and bicarbonates of calcium and magnesium. Hard water can be made soft by boiling or by chemical methods.

Question: C1. Describe the methods to remove permanent hardness of water.

Answer: Permanent hardness is caused by the chlorides and sulphates of calcium and magnesium, which are not removed by boiling. Methods: (i) Adding washing soda Na2CO3, which settles calcium and magnesium as insoluble carbonates at the bottom.

(ii) The sodium zeolite method, in which water is passed through zeolite which exchanges calcium and magnesium for sodium.

(iii) The ion exchange resin method.

Question: C2. Describe the industrial pollutants.

Answer: Industrial pollutants are the waste from factories: chemicals, heavy metals like lead, mercury and cadmium, oil, dyes and hot water. They mix into rivers and land, kill aquatic organisms, poison drinking water and pollute the land.

Question: C3. Prove that water is a solvent.

Answer: Water is called the universal solvent, because it dissolves very many substances. Water is a polar molecule: oxygen carries a partial negative charge and hydrogen carries a partial positive charge. Therefore it dissolves ionic compounds like NaCl and many molecular substances. The hydrogen bonding between water molecules also helps its solvent character.

Question: C4. Describe the disadvantages of hard water.

Answer: Hard water does not form lather with soap, so soap is wasted and clothes are not cleaned. It forms scale in boilers and pipes, which wastes energy and damages machines. Hard water affects the taste and cooking of food. It can also harm the skin and hair.

Question: C5. Describe waterborne diseases in detail.

Answer: Waterborne diseases are caused by drinking or using polluted

water. Cholera and typhoid are caused by bacteria; diarrhoea and dysentery by bacteria or viruses; hepatitis and polio by viruses; hookworm by worms; fluorosis by excess fluoride in water. Prevention: drink clean water, boil or filter water, keep sewage away from drinking water.

7. تجزياتي کيمسٹري

Question: T1. Define analytical chemistry.

Answer: Analytical chemistry is the branch of chemistry that deals with the instruments and methods to separate, identify and quantify matter.

Question: T2. Prove that analytical chemistry is a part of all types of chemistry.

Answer: Analytical chemistry is used in every type of chemistry: in organic, inorganic, physical and biochemistry, analytical methods are needed to identify samples and determine their quantity. Therefore it is a part of all types of chemistry.

Question: T3. Justify that qualifying is the first step to quantify any sample.

Answer: Because first it is necessary to know which compounds are in the sample, and only then can their quantity be determined. If the compound is not identified, the method to measure its quantity cannot be chosen. Therefore first identification, then quantity.

Question: T4. Discuss which method deals with physical properties of substances.

Answer: Physical methods deal with the physical properties of substances, for example atomic emission spectroscopy, Fourier transform spectroscopy, X-ray spectroscopy and trace element analysis. These methods measure the interaction of light with matter.

Question: T5. Differentiate between quantitative and qualitative analysis.

Answer: Qualitative analysis tells which substances are present in a sample, for example chemical and flame tests. Quantitative analysis tells in what quantity those substances are present, for example titrimetric and gravimetric analysis.

Question: T6. Prove that improper functioning of acid-base titration is a systematic error.

Answer: A systematic error is one that occurs in the same way every time and can be estimated and eliminated. Improper functioning in acid-base titration, for example a wrong burette reading or a wrong indicator, produces an error in the same direction in every measurement, so it is a systematic error.

Question: T7. Distinguish between accuracy and precision.

Answer: Accuracy is the agreement between a measured value and the accepted true value. Precision is the agreement between replicate measurements of the same quantity, that is the repeatability of a result. A measurement can be precise but not accurate, accurate but not precise, both or neither.

Question: T8. Explain titration and the apparatus used in titration.

Answer: Titration is the technique in which the concentration of an unknown solution is determined by comparing the volume of a solution of known concentration with the volume of the solution of unknown concentration. Apparatus: burette containing the titrant, flask containing the analyte, pipette, and indicator which shows the end point.

Question: T9. Why is an indicator used in titration?

Answer: An indicator is used because it changes its colour at the end point of the reaction and tells that the titration is complete and the equivalence point has been reached.

Question: T10. Write down the steps of gravimetric analysis.

Answer: (1) Preparation of a solution with known weight of sample.

(2) Separation of the desired constituent.

(3) Weighing the separated constituent.

(4) Computation of the amount of the separated constituent in the sample.

A1. Analytical chemistry deals with instruments and methods to, identify and quantify the matter. Fill in the blank:

(a) Mix

✓ (b) Separate

(c) Differentiate

(d) Manipulate.

A2. The sample may be solid, liquid, gas or a in qualitative analysis. Fill in the blank:

✓ (a) Mixture

(b) Compound

(c) Substance

(d) None of these.

A3. Analysis that deals with the identification of the presence of functional groups in compounds is:

(a) Physical qualitative analysis

(b) Analytical qualitative analysis

✓ (c) Organic qualitative analysis

(d) Inorganic qualitative analysis.

A4. The flame test of copper halide with bluish-green colour identifies the presence of:

✓ (a) Halogen

(b) Hydrogen

(c) Copper

(d) b and c.

A5. The physical methods used to measure physical properties are called:

(a) Combustion analysis method

✓ (b) Atomic emission spectroscopy method

(c) Volumetric analysis method

(d) Gravimetric analysis method.

A6. The error caused by improper functioning of an instrument is:

(a) Determinant error

(b) Indeterminant error

(c) Systematic error

✓ (d) Both a and c.

A7. An agreement between a measured value and the accepted true value is:

(a) Error

✓ (b) Accuracy

(c) Precision

(d) All of these.

A8. Spectroscopy is the interaction of light with:

(a) Liquid

(b) Solid

(c) Gas

✓ (d) Matter.

A9. Gas is the mobile phase in:

(a) Liquid chromatography

(b) Solid chromatography

✓ (c) Gas chromatography

(d) None of these.

A10. It is used to assess the concentration or amount of a given atomic, molecular or ionic chemical:

(a) Chromatography

(b) Spectroscopy

(c) Conductometry

(d) Potentiometry.

Question: B1. What do you think, which method is faster, classical or instrumental?

Answer: The instrumental method is faster. The classical method is slow and needs a large amount of sample; the instrumental method is very fast, sensitive and works with a small amount of sample.

Question: B2. How will you compare the analytical techniques to one another?

Answer: Techniques are compared on the basis of speed, accuracy, precision, amount of sample, cost and training. The classical method is simple and cheap but slow; the instrumental method is fast, sensitive and expensive and needs specialized training.

Question: B3. Can you give some examples of error related to your life?

Answer: Yes. Examples: a wrong weight on a weighing machine; a clock running fast or slow; writing a wrong answer in an exam; measuring a wrong length with a measuring tape; reading a wrong temperature on a thermometer. All of these are errors in measurement.

Question: B4. What do you mean by quantitative analysis?

Answer: Quantitative analysis is the analysis that tells in what quantity a substance is present in a sample. Examples: titrimetric, gravimetric and combustion analysis.

Question: B5. Discriminate which of the following collected volumes of a gas in gas preparation is accurate, precise or accurate and precise both or none of these: First set: 32, 45, 17, 23 cubic centimetre. Second set: 32, 33, 34, 35 cubic centimetre. Third set: 32, 45, 45, 32 cubic centimetre. Fourth set: 32, 32, 33, 32 cubic centimetre. The true value is taken as 32

cubic centimetre, the value repeated in the table.

Answer: First set: none of these, the values are scattered and not close to the true value. Second set: accurate but not precise, the values are close to the true value but not repeatable. Third set: none of these, the values are scattered. Fourth set: accurate and precise both, the values are repeatable and close to the true value.

Question: B6. Why do we use potentiometric analysis in advanced instrumental methods?

Answer: Potentiometric analysis is used to find the concentration of a solute in solution. In it the potential between two electrodes is measured by a voltmeter and there is no flow of current. It is used in the analysis of pollutants in water, pharmaceuticals and drugs, quality control in the food industry and clinical chemistry.

Question: B7. How are scientists using infrared spectroscopy in quality control of different industries?

Answer: Infrared spectroscopy tells the identification and quantity of samples in less time and at less cost, and no polluting chemical is required for it. It is used for the specification of functional groups in food products, polymers and industries, and it is an effective tool for quality control in different industries.

Question: B8. List down the applications of conductometry.

Answer: With conductometry: the degree of dissociation constant can be determined; the solubility of a sparingly soluble salt can be determined; the rate constant of a reaction can be studied; the end point of titration can be determined.

Question: C1. Distinguish between the following: (a) Quantitative analysis and qualitative analysis (b) Titrimetric analysis and gravimetric analysis.

Answer: (a) Qualitative analysis tells which substances are in a sample;

quantitative analysis tells in what quantity they are.

(b) Titrimetric analysis is based on the measurement of volume: the volume of a solution of known concentration is used to find an unknown concentration. Gravimetric analysis is based on the measurement of weight: the analyte is converted into some product and then weighed.

Question: C2. Prove that instrumental analytical methods are more effective than classical analytical methods.

Answer: Instrumental methods are more effective because they are very fast, sensitive, work with a small amount of sample, and give both identification and quantity in less time. Classical methods are slow, need a large amount of sample, and accuracy decreases with decreasing amount. Instrumental methods can also analyse trace amounts and work automatically.

Question: C3. Describe gas chromatography in detail.

Answer: Gas chromatography is an analytical technique for the separation of gases and volatile compounds. The separation takes place by the exchange between a mobile gas phase and a liquid or solid stationary phase. Instruments: gas cylinder, sample injector, gas chromatograph column, detector and data collection device. Gas is the mobile phase; when compounds reach the detector, the detector detects their elution and sends signals to the computer. It is used in the analysis of inorganic compounds, carbohydrates, proteins, lipids, vitamins, pollutants like benzene, plastic materials and dairy products.

Question: C4. Justify that electrochemical methods depend upon electrochemical cells.

Answer: Electrochemical methods measure the potential, charge and electrical quantity of a solution, and this measurement is carried out with

an electrochemical cell. The cell has two half cells, with anode and cathode electrodes, connected by a salt bridge. At the anode electrons are liberated in oxidation and at the cathode electrons are consumed in reduction. When equilibrium comes in both half cells the net voltage becomes zero and the cell stops producing electricity. Therefore these methods depend upon cells.

8. صنعتي کيمسٹري

A1. Soap is the term for salts of a:

(a) Carboxylic acid

(b) Citric acid

(c) Sulphuric acid

✓ (d) Fatty acid.

A2. Surfactants reduce the of water. Fill in the blank:

(a) Viscosity

✓ (b) Surface tension

(c) Boiling point

(d) Melting point.

A3. The carboxylate end of the soap molecule that is attracted to water is called:

(a) Hydrophobic end

(b) End point

✓ (c) Hydrophilic end

(d) None of these.

A4. The use of potassium hydroxide produced a:

(a) Hard soap

✓ (b) Soft soap

(c) Moderate soap

(d) All of these.

A5. Citric acid is used in preparation of cold drinks for:

(a) Sweet taste

(b) Bitter taste

✓ (c) Sour taste

(d) Salty taste.

A6. The centrifuge machine is used for separation of:

(a) Juice

(b) pH

(c) Mud

✓ (d) Crystal.

A7. Abrasives are:

(a) Water soluble minerals

✓ (b) Water insoluble minerals

(c) Water semi soluble minerals

(d) Water absorbing minerals.

A8. Harvesting is the most important step of:

(a) Preparation of soap

(b) Preparation of cold drinks

✓ (c) Preparation of sugar

(d) Preparation of medicines.

A9. Which of the following is used as jet fuel?

✓ (a) Kerosene oil

(b) Diesel oil

(c) Fuel oil

(d) Petrol.

A10. Which one of the following is not a fraction of crude oil?

(a) Paraffin wax

✓ (b) Ammonia

(c) Fuel oil

(d) Petroleum coke.

Question: B1. Define the saponification process.

Answer: Saponification is the chemical reaction in which fat or oil reacts

with caustic soda NaOH or caustic potash KOH to form soap and glycerine.

Question: B2. Describe that NaOH or KOH are used in preparation of soap.

Answer: In soap preparation NaOH or KOH is used as the alkali. NaOH gives hard soap and KOH gives soft soap, because potassium soaps are more soluble in water. The alkali reacts with fat or oil to form soap and glycerine.

Question: B3. List down the raw material needed for sugar preparation.

Answer: The raw material needed for the preparation of sugar is: sugar cane, lime and water.

Question: B4. Explain the components of soft drinks.

Answer: The basis of soft drinks is syrup, made up of water, sugar, acid, coloring and flavoring agents. It includes water, calcium and other minerals, coloring and flavoring agent, sugar for microbial growth, and citric acid for sour taste. The syrup is prepared by dissolving these ingredients into water to 65 Brix.

Question: B5. Define petroleum.

Answer: Petroleum is a natural substance trapped in rocks beneath the earth's crust. The term petroleum refers to rock oil. It is a complex combination of gaseous, liquid and solid hydrocarbons, with water, salts and earth particles also present in it. It is a liquid lighter than water yet insoluble in it.

Question: B6. Justify that petroleum is "Black Gold".

Answer: Petroleum is called Black Gold because it is black and as valuable as gold. From it come petrol, diesel, kerosene, gas, plastics and many other things. It is the main source of the economy and energy of countries, so its value is like that of gold.

Question: C1. Describe the fractions of petroleum in detail.

Answer: Petroleum is separated into fractions by fractional distillation,

which is done in refineries in fractionating columns, cold at the top and hot at the bottom. Fractions: refinery gas -160 to -5 C, carbons 1 to 4, for home heating and cooking; petrol 40 to 110 C, carbons 5 to 8, car fuel; naphtha 110 to 180 C, carbons 8 to 10, for plastics; kerosene 180 to 260 C, carbons 10 to 16, jet aircraft fuel; diesel 260 to 320 C, carbons 16 to 20, fuel for buses and lorries; fuel oil 320 to 400 C, carbons 20 to 50, for industrial heating; bitumen 400 to 600 C, carbons more than 50, for surfacing roads.

Question: C2. Explain the process of preparation of sugar from sugar cane.

Answer: Stages: (1) Harvesting and delivery: sugarcane is harvested in the cooler months of the year and sent to the factory.

(2) Juice extraction: the cane is cut into chips and the juice is extracted in mills.

(3) Clarification: heat, lime and flocculation aids are added to the juice; lime is a suspension of calcium hydroxide which kills enzymes and brings the pH from 5.0 to 6.5 to neutral; mud is separated.

(4) Concentration: the clarified juice is heated in evaporators from 10 to 15 percent sucrose to 55 to 59 percent sucrose.

(5) Crystallization: the syrup is further evaporated in vacuum pans, seed crystals are added, and sugar crystals form.

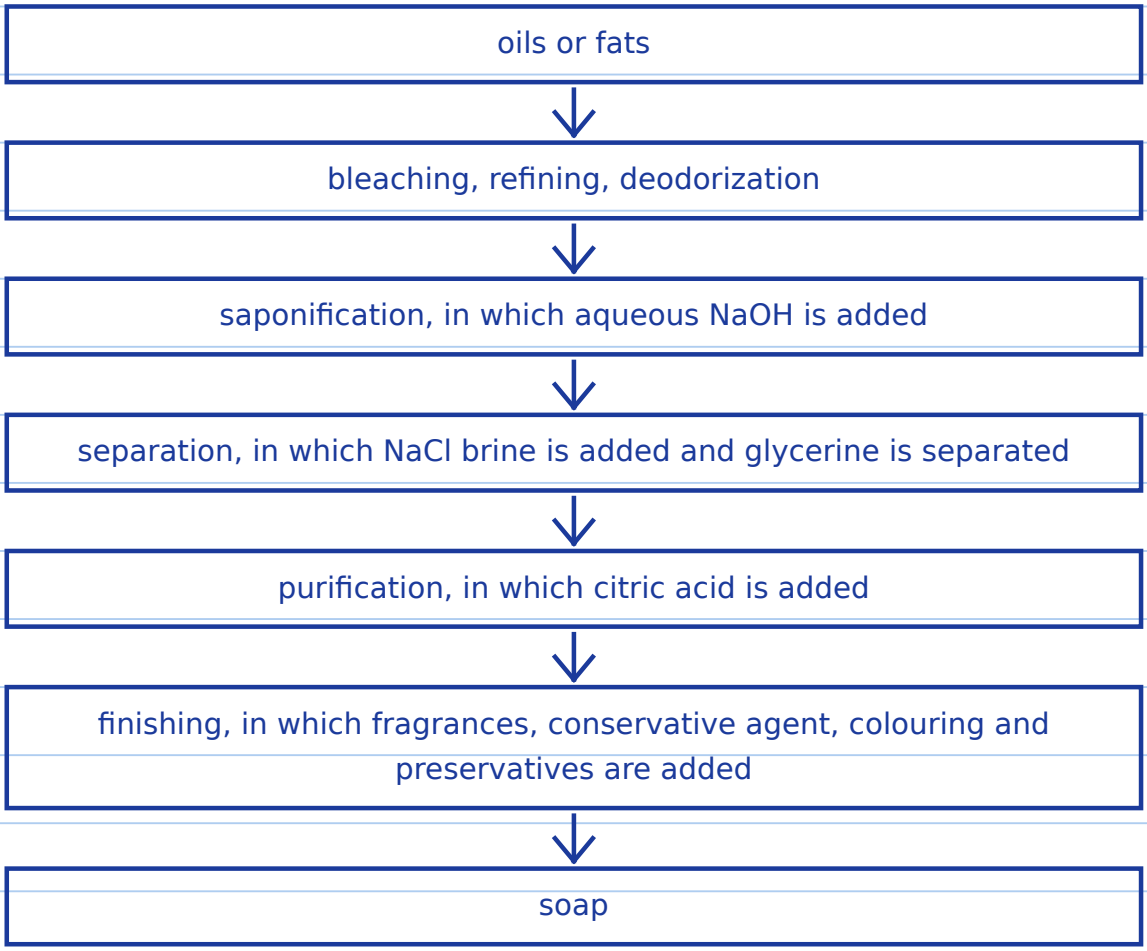
(6) Crystal separation and drying: the crystals are separated from molasses in a basket-type centrifuge machine and dried.

Question: C3. Write down the importance of the pharma industry.

Answer: The pharmaceutical industry studies, manufactures, sells and distributes drugs for the prevention, treatment and cure of illnesses. Importance: treatments increase life expectancy; the objective is to eradicate diseases, for example the elimination of smallpox; pain and suffering are reduced; vaccines save money; hospital stays are shorter; the industry employs millions of people; pharmaceutical companies boost the global economy.

Question: C4. Describe the soap preparation with the help of the flow sheet diagram.

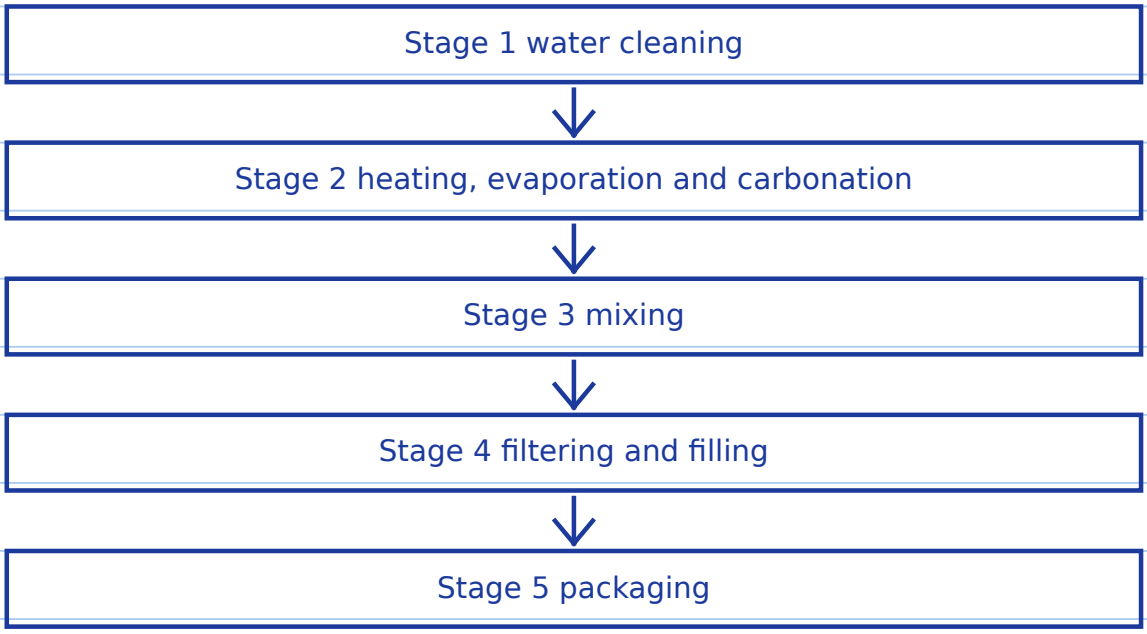
Answer: Stages in order: oils or fats -> bleaching, refining, deodorization -> saponification, in which aqueous NaOH is added -> separation, in which NaCl brine is added and glycerine is separated -> purification, in which citric acid is added -> finishing, in which fragrances, conservative agent, colouring and preservatives are added -> soap.



Flow diagram drawn from the steps given in the answer.

Question: C5. Draw the stepwise preparation of soft drinks in a flow sheet diagram.

Answer: Stages in order: Stage 1 water cleaning: raw water is passed through a filter, water softener, chemicals and pump. Stage 2 heating, evaporation and carbonation: the water is heated with electric heaters, passed through a cooling pipe, and carbon dioxide is added to make carbonated water. Stage 3 mixing: colouring, syrup and flavour are added in the mixing tank. Stage 4 filtering and filling: the drink is filtered and filled into bottles and cans. Stage 5 packaging: packed in cartons and sent to the supermarket.



Flow diagram drawn from the steps given in the answer.